

Self-correcting reasoning loops in mathematical LLMs

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August 12, 2026

Abstract

Systematic Literature Review.

solving process, minimizing unnecessary iterations and maximizing accuracy.

1 Self-correcting Reasoning Loops in Mathematical Large Language Models

2 Original Theoretical Framework

2.1 Hybrid Reasoning Framework

We propose a hybrid reasoning framework that integrates large language models (LLMs) with deterministic classical systems to create self-correcting reasoning loops. This framework leverages the strengths of both approaches to improve the accuracy and reliability of mathematical reasoning tasks.

2.2 Key Components

1. **LLM Component:** Utilize pre-trained LLMs to perform initial reasoning and generate intermediate results.
2. **Classical System Component:** Employ deterministic classical systems to validate and correct the LLM's output, ensuring accuracy and reliability.
3. **Feedback Loop:** Implement a feedback loop to refine the LLM's performance based on the classical system's corrections, enabling continuous improvement.

3 Novel Evaluation Metrics

3.1 Reasoning Accuracy Score (RAS)

To evaluate the performance of self-correcting reasoning loops, we introduce the Reasoning Accuracy Score (RAS). RAS measures the accuracy of the LLM's output in conjunction with the classical system's corrections.

3.2 Reasoning Flow Score (RFS)

We also propose the Reasoning Flow Score (RFS), which assesses the efficiency and effectiveness of the reasoning process. RFS evaluates the LLM's ability to navigate the problem-

4 Limitations

While the hybrid reasoning framework and novel evaluation metrics show promise, there are several limitations to consider:

1. **Scalability:** The classical system component may introduce computational overhead, limiting the scalability of the framework.
2. **Data Quality:** The accuracy of the LLM's output depends on the quality of the training data, which may be affected by biases and errors.
3. **Interpretability:** The hybrid framework may require additional efforts to interpret and understand the reasoning process, potentially hindering explainability.

5 Conclusion

, our proposed hybrid reasoning framework and novel evaluation metrics provide a comprehensive solution for self-correcting reasoning loops in mathematical LLMs. By integrating LLMs with deterministic classical systems and introducing RAS and RFS, we can improve the accuracy, reliability, and efficiency of mathematical reasoning tasks.

6 Future Work

Future research directions include:

1. **Improving Scalability:** Developing more efficient classical system components to enhance the framework's scalability.
2. **Enhancing Data Quality:** Investigating methods to improve data quality and reduce biases in LLM training data.
3. **Increasing Interpretability:** Exploring techniques to enhance the interpretability of the hybrid framework and its reasoning process.

7 References

This manuscript draws inspiration from various research papers, including:

1. Daya Guo et al. (2025) - "DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning"
2. Xu F et al. (2025) - "Toward large reasoning models: A survey of reinforced reasoning with large language models"
3. Mahmoud Masoud et al. (2026) - "Employing in-context learning prompts with large language models for drone routing in delivery services"
4. Michelle Cannon (2025) - "Toward Self-Correcting Hybrid AI Systems: Integrating LLMs with Deterministic Classical Systems"
5. Dinithi Dissanayake et al. (2025) - "Navigating the State of Cognitive Flow: Context-Aware AI Interventions for Effective Reasoning Support"
6. Karan Singhal et al. (2025) - "Large language models encode clinical knowledge"

References